

# EUROPEAN CURRICULUM VITAE

## PERSONAL INFORMATION

Name **Agostino Manzato**



Personal address **via Michelstaedter, 6  
I-34170 Gorizia (GO)  
Italy**

Office address **via Oberdan, 18/a  
I-33040 Visco (UD)  
Italy**

Telephone **+39 0432 934163**

E-mail **agostino.manzato at osmer.fvg.it**

Nationality **Italian**

Civil Status **married**

Date of birth **16.08.1968**

## WORK EXPERIENCE

- Dates (from – to) **01.07.2000 - now**
- Name of organization **Agenzia Regionale per la Protezione dell'Ambiente - Osservatorio Meteorologico Regionale  
Regional Agency for Environmental Protection - Regional Meteorological Observatory**
- Type of organization **Public agency**
- Organization addresses **OSMER – ARPA FVG  
via Oberdan, 18/a  
I-33040 Visco (UD)  
Italy  
Tel. +39 0432 934111  
FAX +39 0432 934100**
- Type of business or sector **Meteorology**
  - Position held **Meteorologist in permanent position**

- Main activities
  - radio-sounding analysis and atmosphere instability evaluation;
  - satellite analysis and instability evaluation of IASI vertical retrievals;
  - thunderstorm, rain and hail forecasting using Neural Networks;
  - linear and non-linear downscaling and statistical adaptation;
  - forecast verification for binary and continuous variables;
  - visualization of meteorological data for real time and case studies.
- Dates (from – to) **20.11.1996 - 30.06.2000**
- Name of organization Centro Servizi Agrometeorologici  
Center for Agrometeorological Services
- Organization addresses CSA  
Via Carso, 3  
I-33052 Cervignano del Friuli (UD)  
Italy
- Type of organization Public agency
- Type of business or sector Meteorology, weather forecasts
  - Position held Meteorologist in permanent position
  - Note I started working in the “*Thunderstorm and Hail Prevention Research Center*”, under the scientific direction of dr. Griffith Morgan. “CSA” became “OSMER – ARPA FVG” on 01.07.2000

## EDUCATION AND TRAINING

- Dates **23.02.1994**
- Name of organization Università degli Studi di Padova  
Padua University - Italy
- Subject Laurea in fisica  
Degree in physics
- Title of qualification awarded Dottore in fisica (104/110)  
Italian university degree in Physics (104/110)
- Further information Thesis titled: *Numerical simulations of the origin of the structures in the large scale universe*, prof. Sabino Matarrese relator and dr. Lauro Moscardini co-relator.

## SCHOOLS AND WORKSHOPS

- Dates **October 1997**
- Name of organization HMZ - EUMETSAT - Ljubljana - Slovenia
- Subject Satellite meteorology and numerical weather prediction workshop
- Dates **April - May 1998**
- Name of organization NSSL, SPC and Oklahoma State University in Norman OK and NCAR in Boulder CO - USA
- Subject I made a one month long "training visit" to these important US Meteorological Centers where I met many researchers (H. Brooks, C. Doswell, D. Zrnica; J. Dye, C. Knight, J. Corbet,...)
- Dates **June 1998**
- Name of organization International Centre for Theoretical Physics - Trieste - Italy
- Subject Summer colloquium on the physics of the weather and climate “The effect of the topography on the atmospheric circulation” (featuring also prof. *F. Mesinger*)

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| Dates                | <b>June 1999</b>                                                                                                                   |
| Name of organization | NCAR, Boulder CO - USA                                                                                                             |
| Subject              | Summer Colloquium on Ice formation in the Atmosphere (featuring also prof. W. Al Cooper)                                           |
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| Dates                | <b>February 2000</b>                                                                                                               |
| Name of organization | Meteo France – Toulouse, France                                                                                                    |
| Subject              | First Conference on European Tornadoes and Severe Storms                                                                           |
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| Dates                | <b>December 2000</b>                                                                                                               |
| Name of organization | ZAMG, EUMETNET, Vienna - Austria                                                                                                   |
| Subject              | First Short-range numerical weather prediction workshop on Statistical Adaptation                                                  |
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| Dates                | <b>June - July 2002</b>                                                                                                            |
| Name of organization | ISAC, CIMA, University of Genoa and Basilicata University, Saint-Oyen (Aosta) - Italy                                              |
| Subject              | Grand Combin Summer School on Physics and Predictability of Rainfall and Floods (featuring also prof. K. Emanuel)                  |
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| Dates                | <b>August 2002</b>                                                                                                                 |
| Name of organization | CHMI, EMS, Prague – Czech Republic                                                                                                 |
| Subject              | Second European Conference on Severe Storms                                                                                        |
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| Dates                | <b>July 2004</b>                                                                                                                   |
| Name of organization | University of Trento, Trento - Italy                                                                                               |
| Subject              | Summer School on Mountain Meteorology "Orographic Effects on Precipitation" (featuring also dr. R. Rotunno)                        |
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| Dates                | <b>November 2004</b>                                                                                                               |
| Name of organization | University of León, INM, EMS, León - Spain                                                                                         |
| Subject              | Third European Conference on Severe Storms,                                                                                        |
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| Dates                | <b>June 2005</b>                                                                                                                   |
| Name of organization | ZAMG, EUMETNET, Vienna - Austria                                                                                                   |
| Subject              | Third Short-range numerical weather prediction workshop on Statistical Adaptation                                                  |
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| Dates                | <b>July 2005</b>                                                                                                                   |
| Name of organization | CISM, OSMER, Udine - Italy                                                                                                         |
| Subject              | CISM Advanced School on Atmospheric Convection: Research and Operational Forecasting Aspects (featuring also prof. P. Markowsky)   |
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| Dates                | <b>January 2006</b>                                                                                                                |
| Name of organization | American Meteorological Society, Atlanta (GE) - USA                                                                                |
| Subject              | Short Course on Artificial Intelligence Applications to Environmental Science (featuring also prof. C. Marzban)                    |
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| Dates                | <b>January 2006</b>                                                                                                                |
| Name of organization | American Meteorological Society, Atlanta (GE) - USA                                                                                |
| Subject              | 86th AMS Annual Meeting                                                                                                            |
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| Dates                | <b>September 2007</b>                                                                                                              |
| Name of organization | EUMETSAT, ICTP, Trieste - Italy                                                                                                    |
| Subject              | EUMETSAT Training Workshop on Interpretation of Satellite Images/Products of Convective Storms (featuring also prof. D. Rosenfeld) |
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| Dates                | <b>September 2007</b>                                                                                                              |
| Name of organization | OSMER, ICTP, EMS, Trieste - Italy                                                                                                  |
| Subject              | Fourth European Conference on Severe Storms                                                                                        |

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| Dates                | <b>May 2008</b>                                                                                                                                                                |
| Name of organization | SAR, ARPA Sardegna, ISAC, Alghero - Italy                                                                                                                                      |
| Subject              | Third Mediterranean School on Mesoscale Meteorology: "Fundamentals and applications of probabilistic forecasts and Ensemble Prediction Systems" (featuring also dr. T. Hamill) |
| Dates                | <b>November 2008</b>                                                                                                                                                           |
| Name of organization | ICTP, Trieste - Italy                                                                                                                                                          |
| Subject              | Conference on Teleconnections in the Atmosphere and Oceans                                                                                                                     |
| Dates                | <b>October 2009</b>                                                                                                                                                            |
| Name of organization | EUMETSAT, Landshut - Germany                                                                                                                                                   |
| Subject              | Convection Working Group Workshop                                                                                                                                              |
| Dates                | <b>October 2009</b>                                                                                                                                                            |
| Name of organization | ESSL, Landshut - Germany                                                                                                                                                       |
| Subject              | Fifth European Conference on Severe Storms                                                                                                                                     |
| Dates                | <b>June 2010</b>                                                                                                                                                               |
| Name of organization | ISAC - CNR, Bologna - Italy                                                                                                                                                    |
| Subject              | Fourth HyMeX workshop                                                                                                                                                          |
| Dates                | <b>July 2010</b>                                                                                                                                                               |
| Name of organization | ZAMG, Wien - Austria                                                                                                                                                           |
| Subject              | INCA-CE Workshop                                                                                                                                                               |
| Dates                | <b>March 2011</b>                                                                                                                                                              |
| Name of organization | EUMETSAT, Darmstadt - Germany                                                                                                                                                  |
| Subject              | Evaluating atmospheric instability from high spectral resolution IR satellite observations (featuring dr. P. Antonelli)                                                        |
| Dates                | <b>October 2011</b>                                                                                                                                                            |
| Name of organization | Baleari University, Palma de Mallorca - Spain                                                                                                                                  |
| Subject              | Sixth European Conference on Severe Storms                                                                                                                                     |
| Dates                | <b>March 2012</b>                                                                                                                                                              |
| Name of organization | EUMETSAT, Prague – Czech Republic                                                                                                                                              |
| Subject              | Convection Working Group Meeting                                                                                                                                               |
| Dates                | <b>September 2012</b>                                                                                                                                                          |
| Name of organization | CNR-ISAC, Castro Marina – Italy                                                                                                                                                |
| Subject              | Fourth ISAC Summer School: "Severe Convection Weather: Theory and Applications" (featuring also prof. E. Fedorovich)                                                           |
| Dates                | <b>April 2013</b>                                                                                                                                                              |
| Name of organization | OSMER – ZAMG, Grado - Italy                                                                                                                                                    |
| Subject              | Sixth INCA-CE meeting and EUMETNET-Nowcasting Activity joint session (featuring also dr. P. Nurmi)                                                                             |
| Dates                | <b>June 2013</b>                                                                                                                                                               |
| Name of organization | Univ. Ljubljana – ARSO, Kranjska Gora - Slovenia                                                                                                                               |
| Subject              | 32nd ICAM International Conference on Alpine Meteorology                                                                                                                       |
| Dates                | <b>August 2013</b>                                                                                                                                                             |
| Name of organization | CISM – ISAC, Udine - Italy                                                                                                                                                     |
| Subject              | The Fluid Dynamics of Climate (featuring also prof. M. Ghil)                                                                                                                   |
| Dates                | <b>March 2014</b>                                                                                                                                                              |
| Name of organization | OSMER – ISAC, Visco - Italy                                                                                                                                                    |
| Subject              | Previsioni di Limited Area Model in casi di Severe Weather (featuring also dr M. M. Miglietta)                                                                                 |

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| Dates                | <b>April 2014</b>                 |
| Name of organization | EUMETSAT – DMHZ, Zagreb - Croatia |
| Subject              | Convection Working Group Meeting  |

  

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| Dates                | <b>April 2014</b>                    |
| Name of organization | ZAMG, Vienna – Austria               |
| Subject              | First European Nowcasting Conference |

  

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| Dates                | <b>April-May 2014</b>                        |
| Name of organization | European Geosciences Union, Vienna – Austria |
| Subject              | EGU conference                               |

  

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| Dates                | <b>June 2014</b>                            |
| Name of organization | Bern University, Bern – Swiss               |
| Subject              | First European Hail Workshop (invited talk) |

  

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| Dates                | <b>September 2015</b>                                    |
| Name of organization | Innsbruck University, Innsbruck – Austria                |
| Subject              | 33rd ICAM International Conference on Alpine Meteorology |

  

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| Dates                | <b>September 2015</b>                       |
| Name of organization | ESSL, Wiener Neustadt – Austria             |
| Subject              | Eighth European Conference on Severe Storms |

  

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| Dates                | <b>April 2016</b>                                 |
| Name of organization | EUMETSAT – Aeronautica Militare, Florence – Italy |
| Subject              | Convection Working Group Meeting                  |

  

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| Dates                | <b>June 2016</b>                                                                                                                |
| Name of organization | CNR-ISAC, Castro Marina – Italy                                                                                                 |
| Subject              | Sixth ISAC Summer School: “Advances in severe weather analysis: models and observations”<br>(featuring also prof. H. Bluestein) |

## PERSONAL SKILLS AND COMPETENCES

MOTHER TONGUE **ITALIAN**

### OTHER LANGUAGES

#### **ENGLISH**

- Reading skills Very good
- Writing skills Very good
- Verbal skills Good

#### **SLOVENIAN**

- Reading skills Goodt
- Writing skills Sufficient
- Verbal skills Sufficient

ORGANIZATIONAL SKILLS,  
REVIEW AND MEMBERSHIP

In September 2002 I have organized Seminar on Orographic Precipitations with dr. R. Rotunno and prof. R. Steinacker at OSMER; in March 2014 I have organized a Seminar on Severe Weather Local Area Model forecasts with dr. Mario Marcello Miglietta and Silvio Davolio at OSMER.

I have made reviews of scientific papers for many journals, including also *Journal of Geophysical Research*, *Atmospheric Research*, *Atmospheric Science Letters*, *Journal of Applied Meteorology and Climatology*, *Journal of the Atmospheric Sciences*, *Quarterly Journal of the Royal Meteorological Society*, *Meteorological Applications*, *Theoretical and Applied Climatology* and *International Journal of Climatology*.

I'm member of the *Convective Working Group* and of the *EUMETNET-NOWCASTING Activity group*.

TEACHING EXPERIENCE

In 2003-2005 I was lecturer for the Applied Meteorology course at the University of Udine - Italy in the frame of the Computational Physics laurea (Master degree).

In 2006 I was tutor of a student during his thesis work in the frame of the master degree in Mathematics at the University of Udine - Italy.

On 17/11/2015 I was invited by ARPA Veneto to hold a seminar on atmosphere instability and sounding-derived instability indices in Teolo (PD).

TECHNICAL SKILLS  
AND COMPETENCES

Computer knowledge at good level (like system administrator):

Linux (RedHat, SuSe, Ubuntu), Unix (SGI Irix), Windows (98, XP, Vista, 7, 10).

Programming languages at good level:

Python, Shell Script (BASH, TCSH), R (statistical language). I used to program in Fortran and C many years ago.

Software tools and packages:

ZEBRA (NCAR visualization tool for GRIB and NetCDF data), SatSignal (for MeteoSat and MSG data), MSG\_RGB, HTDIG (search engine), LaTeX, OpenOffice, GIMP, PhotoShop.

I have experience on stepwise regression and Artificial Intelligence techniques (Neural Networks, regression trees,...) applied to meteorological forecast problems.

I manage the data format conversion in GRIB (using Lats library) and NetCDF.

I manage the OSMER internal Intranet. I'm a skilled photographer (analog and digital).

## SELECTED PUBLICATIONS

Corresponding author:

- Manzato, A., Cicogna, A., and A. Pucillo, 2016: *6-hour maximum rain in Friuli Venezia Giulia: climatology and ECMWF-based forecasts*, Atmos. Res. 169 B, 465-484.
- Feudale, L. and A. Manzato, 2014: *Cloud-to-Ground Lightning Distribution and its Relationship with Orography and anthropogenic emissions in the Po Valley*, Journal of Applied Meteorology and Climatology, 53, 2651-2670.
- Manzato, A., Davolio, S., Miglietta, M. M., Pucillo, A., and M. Setvák, 2014: *12 September 2012: A supercell outbreak in NE Italy?* Atmos. Res., 153, 98-118.
- Manzato, A., 2013: *Hail in NE Italy: A neural network ensemble forecast using sounding-derived indices*, Weather and Forecasting, 28, 3-28.
- Manzato, A., 2012: *Hail in NE Italy: Climatology and bivariate analysis with the sounding-derived indices*, Journal of Applied Meteorology and Climatology, 51, 449-467.
- Manzato, A., 2008: *A Verification of Numerical Model Forecasts for Sounding-Derived Indices above Udine, NE Italy*, Weather and Forecasting, 23, n3 477-495.
- Manzato, A., 2007: *A Note on the Maximum Peirce Skill Score*, Weather and Forecasting, 22, n5, 1148-1154.
- Manzato, A., 2007: *The 6 Hours Climatology of Thunderstorms and Rainfalls in the Friuli Venezia Giulia Plain*, Atmos. Res. ECSS 2004 special issue, 83, 336-348.
- Manzato, A., 2005: *The use of sounding derived indices for a neural network short-term thunderstorm forecast*, Weather and Forecasting, 20, n6, 896-917.
- Manzato, A. and G. M. Morgan, 2003: *Evaluating the sounding instability with the Lifted Parcel Theory*, Atmos. Res., 67-68, 455-473.

Not lead authored:

- Miglietta, M., A. Manzato, R. Rotunno, 2016: *Characteristics and Predictability of a Supercell during HyMeX SOP1*, Quarterly Journal of RMS, in press
- Mecikalski, J., D. Rosenfeld, A. Manzato, 2016: *Evaluation of Geostationary Satellite Observations and the Development of a 1-2 hour Prediction Model for Future Storm Intensity*, JGR, in press
- Davolio, S., A. Volontè, A. Manzato, A. Pucillo, A. Cicogna, M. E. Ferrario, 2015: *Mechanisms producing different precipitation patterns over North-Eastern Italy: insights from HyMeX-SOP1 and previous events*, Quarterly Journal of RMS, DOI: 10.1002/qj.2731.

## FURTHER DETAILS OF THE CURRENT ACTIVITY

Forecasting with sounding-derived indices and Artificial Neural Network:

I have used the sounding-derived indices as input of ANN ensemble for short-range forecast of thunderstorm and rain occurrence and intensity. Recently I have applied the same methodology to ECMWF model-derived pseudo-indices and also to the hail problem. I have evaluated also the thunderstorm forecasting skill of a system based on IASI satellite-derived data.

Downscaling:

I have supervised a mathematic student during his thesis on downscaling of maximum and minimum daily temperatures from ECMWF data using a linear multiregression (MOS).

Forecast Verification:

I'm particularly interested in verification of the binary (or multi-categorical) classifiers using the Signal Detection Theory and in particular the maximum Pierce Skill Score. I'm also interested in combining classifiers in ensemble (bagging, boosting, MV,...).

## FURTHER INTERESTS

Classical music, Hi-Fi, electronics, photography.

## ADDITIONAL INFORMATION

I have a 8-year old daughter.

## DATE

28 June 2016

**SIGNATURE**

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